

# Appendix H - Per-CPU Symbol Index

Entry points, ABI conventions, and reserved memory regions for each CPU. The detailed register descriptions and the per-CPU chapter give the full story; this appendix is the cheat sheet.

## H.1 IE64

| Symbol            | Value / role                                                                                                           |
|-------------------|------------------------------------------------------------------------------------------------------------------------|
| Reset vector      | \$000000 (first instruction at start of RAM).                                                                          |
| Trap vector base  | \$000400 (8-byte entries, indexed by trap number).                                                                     |
| Supervisor stack  | grows down from \$0A0000.                                                                                              |
| User stack (R31)  | grows down from BASIC's per-program stack region.                                                                      |
| Call ABI          | Arguments R8-R15; result R8; caller-saved R1-R7; callee-saved R16-R30; R0 = 0; R31 = SP.                               |
| FPU regs          | F0-F15; FP32 values, with double operations using register pairs. F0-F7 are argument / result registers by convention. |
| BASIC text        | \$023000-\$04FFFF (BASIC_PROG_START-BASIC_PROG_LIMIT - 1).                                                             |
| Simple vars       | \$050000-\$057FFF.                                                                                                     |
| String vars       | \$058000-\$05FFFF.                                                                                                     |
| Arrays            | \$060000-\$08BFFF.                                                                                                     |
| String temps      | \$08C000-\$08FFFF.                                                                                                     |
| GOSUB / FOR stack | \$090000-\$096FFF.                                                                                                     |

## H.2 IE32

| Symbol       | Value / role                                                                         |
|--------------|--------------------------------------------------------------------------------------|
| Reset vector | \$000000.                                                                            |
| Stack base   | \$09F000 (STACK_START); grows down.                                                  |
| Timing       | WAIT n for short delays; use device status or interrupts for frame and audio timing. |
| Call ABI     | Arguments A,X,Y,Z; result A; B-W caller-saved; stack via PUSH / POP.                 |

## H.3 6502

| Symbol           | Value / role                  |
|------------------|-------------------------------|
| Reset vector     | \$FFFC (low) / \$FFFD (high). |
| IRQ / BRK vector | \$FFFE / \$FFFF.              |
| NMI vector       | \$FFFA / \$FFFB.              |

| Symbol         | Value / role                            |
|----------------|-----------------------------------------|
| Stack page     | \$0100-\$01FF, indexed by S.            |
| Zero page      | \$0000-\$00FF.                          |
| Bank registers | \$F700-\$F705, \$F7F0.                  |
| MMIO aperture  | \$F000-\$FFF9, mirrors \$F0000-\$F0FF9. |
| VGA C64-style  | \$D700-\$D70D.                          |
| ULA paged port | \$D800-\$D817.                          |
| PSG / SID      | \$D400-\$D40F, \$D500-\$D55F.           |
| POKEY          | \$D200-\$D20A.                          |
| TED audio      | \$D600-\$D605.                          |

## H.4 Z80

| Symbol           | Value / role                                          |
|------------------|-------------------------------------------------------|
| Reset vector     | \$0000.                                               |
| NMI vector       | \$0066.                                               |
| RST n            | n * 8 (\$00, \$08, ..., \$38).                        |
| IM 2 vector base | (I << 8)   (data byte from device).                   |
| Bank registers   | port-mapped through bus translation at \$F700-\$F705. |
| MMIO aperture    | \$F000-\$FFF9.                                        |
| PSG / AY ports   | \$F0 select, \$F1 data.                               |
| TED audio ports  | \$F2, \$F3.                                           |
| POKEY ports      | \$D0, \$D1.                                           |
| SID ports        | \$E0, \$E1.                                           |
| SN76489 ports    | \$E4 data, \$E5 status.                               |
| VGA ports        | \$A0-\$AD.                                            |

## H.5 M68K (MC68020-Class)

| Symbol        | Value / role                                          |
|---------------|-------------------------------------------------------|
| Reset vector  | \$0000.0000 (initial SSP) / \$0000.0004 (initial PC). |
| Vector table  | \$0000.0000-\$0000.03FC (256 entries, 4 bytes each).  |
| Bus error     | vector 2.                                             |
| Address error | vector 3.                                             |
| Illegal       | vector 4.                                             |
| Zero divide   | vector 5.                                             |

| Symbol              | Value / role                                                            |
|---------------------|-------------------------------------------------------------------------|
| CHK                 | vector 6.                                                               |
| Trapv               | vector 7.                                                               |
| Privilege violation | vector 8.                                                               |
| Trace               | vector 9 (trace bits are stored; this chip does not raise trace traps). |
| Line A              | vector 10.                                                              |
| Line F              | vector 11.                                                              |
| TRAP #n             | vectors 32-47.                                                          |
| Auto-vector IRQs    | vectors 25-31.                                                          |
| Call ABI            | Arguments on stack; D0 / A0 caller-saved; D2-D7 / A2-A6 callee-saved.   |

## H.6 x86 (8086 + 386 extensions, real-mode only)

| Symbol          | Value / role                                                                                                |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| Reset vector    | EIP = 0, CS = 0, DS = ES = SS = 0 (flat, not the 8086 F000:F000 boot vector).                               |
| IVT             | \$0000-\$03FF (256 entries, 4 bytes each: offset + segment).                                                |
| Stack           | SS:ESP, segments are zero so the stack lives in flat RAM.                                                   |
| Call ABI        | Caller pushes arguments right-to-left; EAX returns; EBX, ESI, EDI, EBP callee-saved; ECX, EDX caller-saved. |
| BIOS-style ints | reserved; no BIOS ROM is provided. The IVT is initialised to a default IRET routine.                        |

Real-mode 20-bit physical address calculation ( $\text{seg} \ll 4 + \text{ofs}$ ) is part of the CPU address path. The 32-bit linear form (the result of the calculation) is what reaches the bus. Programs that use 32-bit immediate addressing reach the full flat address space directly.

## H.7 Cross-CPU bus addresses (shared)

These addresses are the same in every CPU's 32-bit view of the bus. The 8-bit CPUs reach them through the bank-window mechanism described in Chapters 26 and 27.

| Address | Meaning               |
|---------|-----------------------|
| \$F0700 | TERM_OUT.             |
| \$F1400 | HOST appliance block. |
| \$F2200 | File I/O block.       |
| \$F2300 | Media loader.         |
| \$F2320 | RUN loader block.     |
| \$F2340 | Coprocessor.          |
| \$F2400 | SysInfo.              |
| \$F8000 | Voodoo 3D.            |